JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 87, NO. B5, PAGES 3819–3830, MAY 10, 1982

Rotation of the Southernmost Sierra Nevada, California

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Structural, paleogeographic, and magmatic trends of the Sierra Nevada batholith swing from a characteristic NNW trend in the northern and central parts of the batholith to a nearly E-W trend in the Tehachapi and San Emigdio mountains. Paleomagnetic results from eight sites in the upper Cretaceous (80–86 m.y.) Bear Valley Springs pluton in the Tehachapi Mountains yield a mean direction D=022°, I=56° (k=97, α_{95}=5.6°). This mean direction is significantly different from the expected late Cretaceous field direction for this locality and suggests that the pluton has undergone a clockwise rotation of 45°±14°, in accord with the change in structural trend in that area. Lower to middle Miocene volcanic and sedimentary rocks of the Kinnick Formation overlie the plutonic rocks and yield a mean direction D=354°, I=64°. A paleomagnetic pole derived from this mean direction is not significantly different from the expected Miocene pole for North America. This constrains the time of rotation of the pluton to be between about 80 and 20 m.y., probably predating establishment of the present San Andreas transform in this area. We suggest that the observed rotation is part of a large oroclinal bend affecting to different degrees the entire southern Sierra Nevada. The driving mechanism for such a bending may be right-lateral shear along a proto-San Andreas transform, for which there is now a significant body of evidence.

INTRODUCTION

In the southern part of the Sierra Nevada batholith, trends of several structural, paleogeographic, and magmatic elements swing around from the dominant NNW trend of the northern and central parts of the batholith to a nearly E-W trend in the Tehachapi and San Emigdio mountains. Among the deflected elements are the outcrop pattern and western limit in the subsurface of the Sierra Nevada batholith, the belt of upper Cretaceous plutons in the Sierra Nevada [*Evernden and Kistler*, 1970], the quartz diorite line of *Moore* [1959], the line of $^{87}Sr/^{86}Sr$ = 0.706 [*Kistler and Peterman*, 1973], a paleocontinental margin represented by the boundary between metasedimentary terranes of continental and oceanic affinities [*Ross*, 1980] and the trends of septa of these terranes, and a fault zone of complex history comprising the Breckenridge, Kern Canyon, and White Wolf faults. Some of the features show a smooth, regular change in trend; others appear to change trend abruptly.

The change in trend may be an original feature of the batholith and continental margin or the result of later deformation by (1) motion on closely spaced fault planes without rotation between faults, (2) ductile oroclinal bending, (3) rotation of individual fault-bounded blocks, or a combination of these processes. Processes 2 or 3 would be expected to produce anomalous paleomagnetic directions in rocks from the southernmost Sierra Nevada. *Locke et al.* [1940] noted the bend in outcrop pattern and attributed it to right-lateral shear along what is now called the San Andreas fault.

Several recent tectonic syntheses address the possibility of rotation in the southernmost Sierra Nevada. *Burchfiel and Davis* [1981] find evidence for a clockwise oroclinal bend in their early Miocene tectonic reconstruction, based on deflection of pre-Eocene paleogeographical and structural trends. *Luyendyk et al.* [1980] postulate late Oligocene through late Miocene clockwise rotation of several fault-bounded crustal blocks in southern California, among them the Tehachapi Mountains block. *Ross* [1980] discusses the possibility of a bend affecting metasedimentary and intrusive rocks in the southernmost Sierra Nevada and concludes that if present, the bend is sharp and complex rather than being a smooth change. The present study was undertaken to determine whether rotation is recorded in rocks of the Tehachapi area and if so, when it occurred.

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Paper number 1B1838.
0148-0227/82/001B-1838$05.00

GEOLOGIC SUMMARY

The structural setting of the southernmost Sierra Nevada is complex. The region is near the junctions of several structural-tectonic provinces: Sierra Nevada–Great Valley block, Mojave block, Basin and Range province, Salinian block, and Transverse Ranges (Figure 1). Unlike the northern and central parts of the Sierran batholith, which are nearly devoid of major faults, the southern part is cut by numerous faults and, at least in late Cenozoic time, has probably experienced some internal deformation rather than behaving as a rigid block. This region of the Sierra Nevada is bounded on the south by the Garlock fault and on the west by the San Andreas fault and is cut by the Pleito thrust and the Kern Canyon, Edison, White Wolf, Breckenridge, and Sierra Nevada faults.

The basement of the southernmost Sierra Nevada comprises metamorphic terranes of both continental and oceanic affinities intruded by upper Cretaceous tonalite, granodiorite, and granite plutons and minor mafic intrusives of the Sierra Nevada magmatic arc [*Ross*, 1980]. Middle and upper Tertiary sedimentary and volcanic rocks unconformably overlie basement rocks. Samples were collected from the upper Cretaceous Bear Valley Springs (BVS) pluton, the

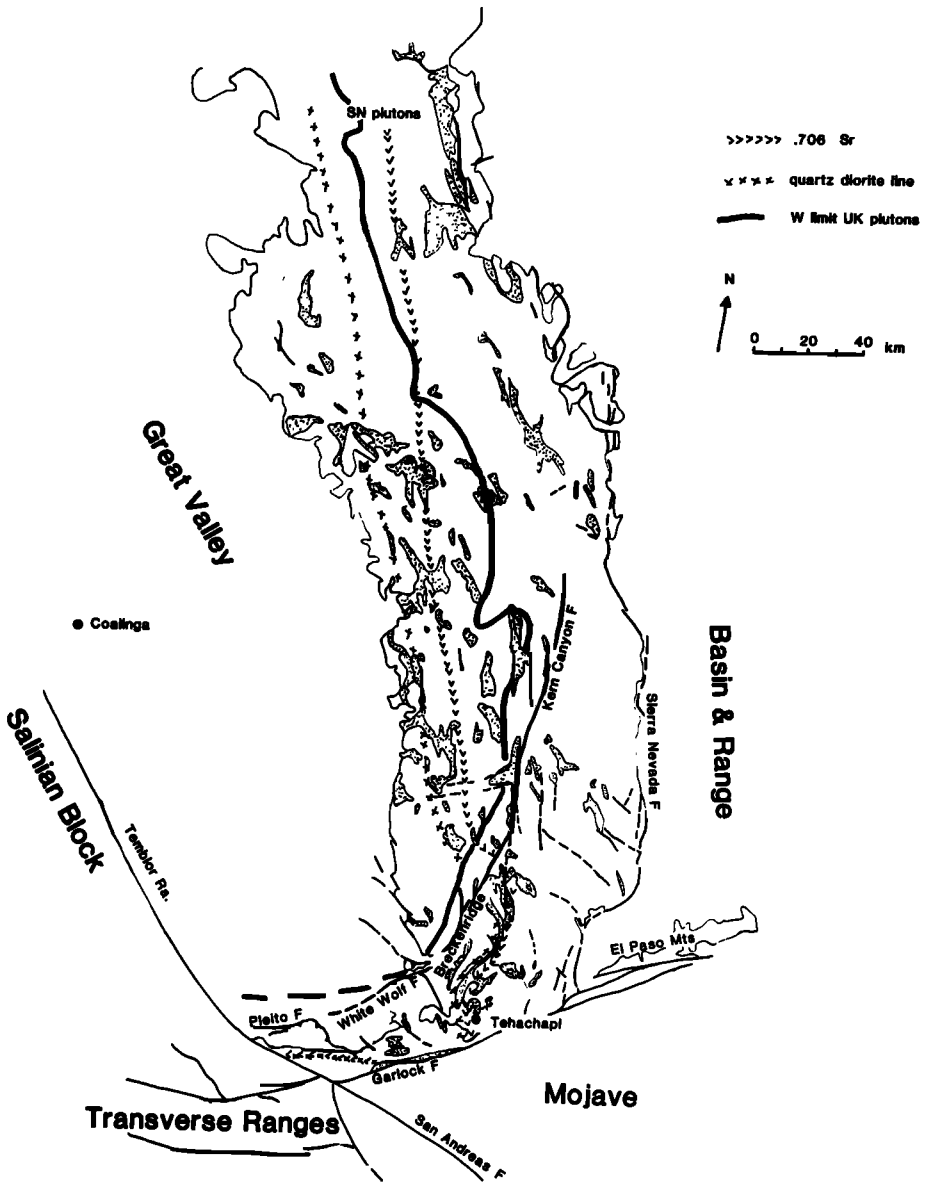


Fig. 1. Central and southern Sierra Nevada batholith (outlined) showing metasedimentary rocks (stippled), faults bounding tectonic provinces and localities mentioned in text. Deflected trends shown schematically are $^{87}Sr/^{86}Sr > 0.706$ [*Kistler and Peterman*, 1973], quartz diorite line [*Moore*, 1959], and the western limit of the belt of upper cretaceous plutons [*Evernden and Kistler*, 1970].

Miocene Kinnick Formation, and from unnamed volcanic rocks in the vicinity of the town of Tehachapi (Figure 2). The geology of the sample sites is described by *Ross* [1980], *Dibblee and Louke* [1970], *Samsel* [1962], and *Dibblee and Chesterman* [1953].

If the southern Sierra Nevada is viewed as a large-scale, simple 'hook-shaped' structure, the study site is about midway around the hook. Looking at the smaller-scale pattern of metasedimentary rocks [*Ross*, 1980], the site is at the southern end of the north trending domain rather than in the east trending domain.

The Bear Valley Springs pluton is a medium-grained, hornblende-biotite tonalite with abundant flattened or elongate inclusions and dark patches which probably represent partially digested country rock. It has a moderately developed foliation resulting from alignment of inclusions and mafic minerals. *Ross* [1980] believes that the pluton probably assimilated and replaced a great deal of metamorphic material, as a very diffuse and gradational contact zone is found between the BVS pluton and an amphibolite gneiss to the south. The locations of two of the radiometric age determinations on the BVS pluton are shown on Figure 2. K/Ar ages are 81.2 m.y. from biotite [*Evernden and Kistler*, 1970], 85.9±2.6 m.y. from biotite and 88.1±2.6 m.y. from hornblende from a single sample (J. Morton, personal communication, 1980). Whole rock Rb/Sr isochron ages are 80 m.y. and 81 m.y. [*Kistler and Peterman*, 1978]. The BVS pluton is intruded by the upper Cretaceous granodiorite of Claraville and the tonalite of Mount Adelaide [*Ross*, 1980].

Tertiary rocks of the Tehachapi area are described by *Dibblee and Louke* [1970] and summarized below. The oldest Tertiary unit, the Witnet Formation, is composed of

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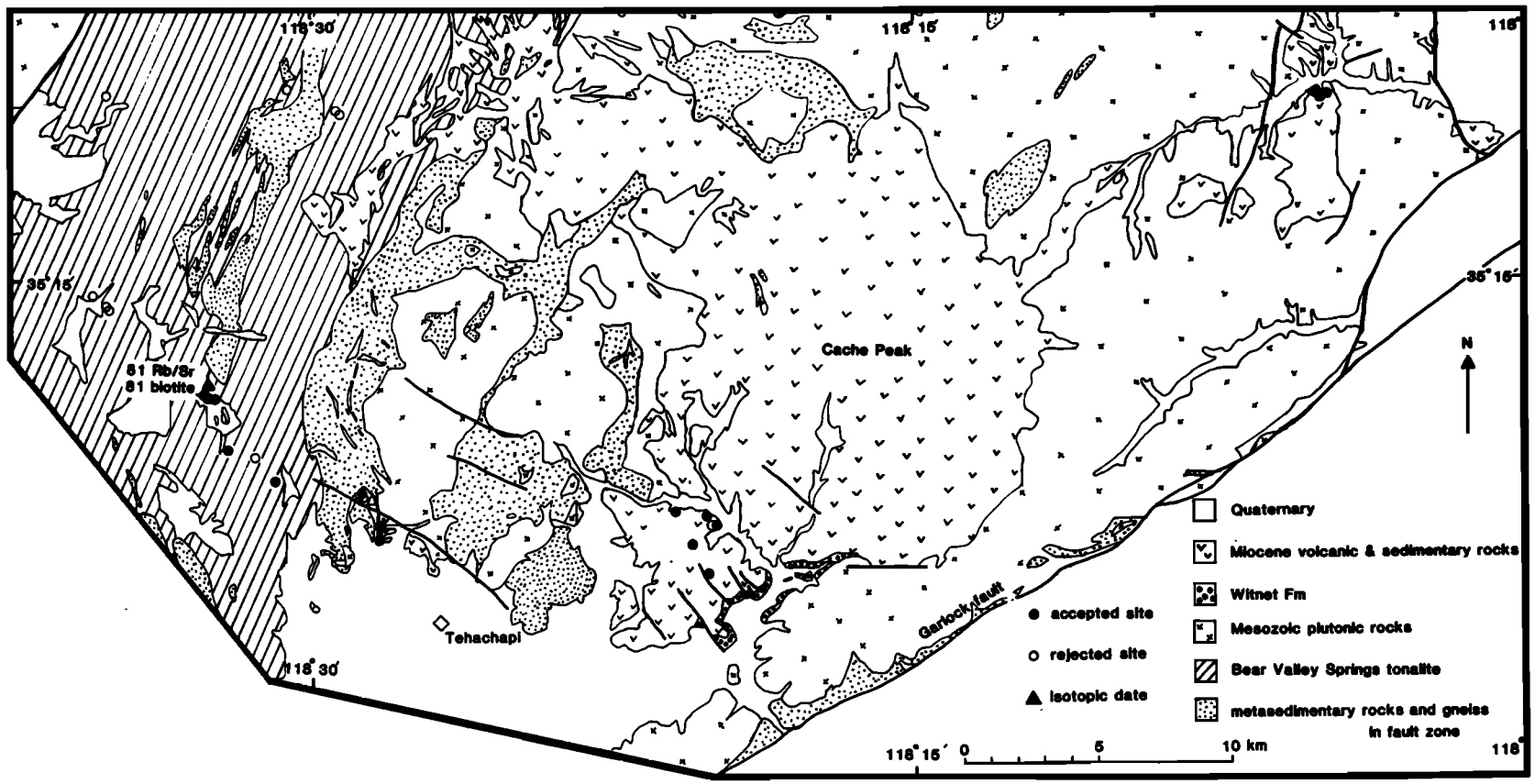


Fig. 2. Generalized geologic map of area of study after *Ross* [1980], *Dibblee and Louke* [1970], *Dibblee and Warne* [1970], *Samsel* [1962], and *Dibblee and Chesterman* [1953]. Sample sites and locality of two age determinations shown.

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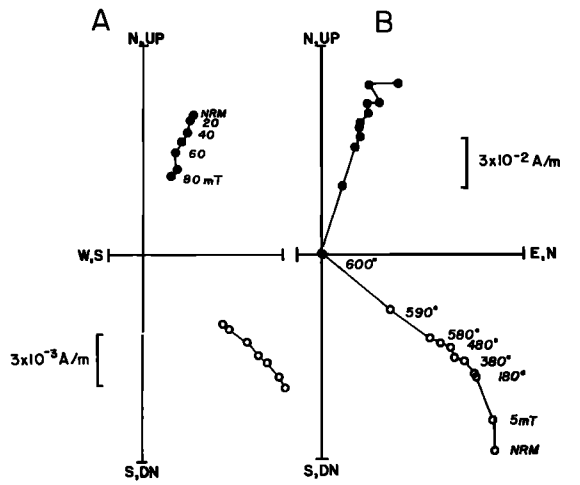


Fig. 3. Typical orthogonal vector plots for BVS pluton. (*a*) AF demagnetization (mT). (*b*) Thermal demagnetization (°C). Solid circles represent directions on a horizontal plane; open circles represent directions on a vertical, north-south plane.

nonmarine conglomerate, sandstone, and siltstone and is probably correlative with the Paleocene Goler Formation of the El Paso Mountains. It may have been deposited in a restricted fault trough along the site of the present Garlock fault (B. Cox, personal communication, 1980). The Witnet Formation is now present only in a narrow band close to the Garlock fault and is overlain with angular unconformity by Miocene rocks. The BVS pluton and most other plutonic rocks in the region are directly overlain buttress-fashion by Miocene rocks. The lowest Miocene unit is the Kinnick Formation, made up of tuffaceous pyroclastics and lacustrine marls, some of which are silicified. Hemingfordian mammal fossils date the upper part of the formation as lower to middle Miocene [*Dibblee and Louke*, 1970]. The Kinnick is conformably overlain by arkosic sandstone and siltstone of the Bopesta Formation. Intercalated in and intertonguing with the upper Kinnick and lower Bopesta formations are flows of andesite, basalt, and rhyolite whose source is a volcanic center just to the northeast in the Cache Peak–Emerald Mountain region (Figure 2).

PALEOMAGNETIC STUDY

In the BVS pluton, 82 samples were collected from 16 sites. Seventy four samples of Miocene basalt and sedimentary rocks were collected at 16 sites (Figure 2). Samples were taken from roadcuts and stream canyons as field-drilled cores and occasionally as oriented blocks. A sun compass was used to orient plutonic and volcanic samples. Accuracy of orientation is estimated to be ≤2° for field-drilled cores and 3°–4° for block samples.

Remanence measurements were made on a two-axis superconducting magnetometer. All specimens were progressively demagnetized by AF (alternating field) and thermal methods. AF demagnetizations were done inside the magnetometer using a two-coil system. Thermal demagnetization used a Schonstedt TSD-1 shielded furnace apparatus. After thermal demagnetizations specimens were demagnetized at 5 mT before remeasuring to remove any viscous magnetization acquired during transfer from furnace to magnetometer. A least squares technique similar to that of *Kirschvink* [1980] was used to isolate mean magnetization directions from vector trajectories.

PALEOMAGNETIC RESULTS—BVS PLUTON

As is frequently the case for plutonic rocks [e.g., *Teissere and Beck*, 1973; *Beske et al.*, 1973; *Wilson and Cox*, 1980], stability of magnetization in the BVS pluton is variable. In general, magnetizations of these specimens are either highly stable with similar directions or highly unstable. With the exception of two sites, all rocks of the same lithology within a given site show similar behavior. Unstable behavior is marked by failure to yield a consistent magnetic direction on repeated demagnetization at the same peak field or erratic changes of direction and intensity upon progressive thermal or AF demagnetization. Stable directions very different from the mean magnetic direction for the pluton are not observed except in one of the two partially stable sites just mentioned. This site (site K) was ultimately rejected because no samples had similar directions. The other partially stable site, site L, was included in calculation of the mean direction, although it includes only three samples and has a large α_{95}. Altogether, 9 sites were judged unstable and were rejected (Figure 2). Orthogonal vector plots representative of the sites with stable magnetizations are shown in Figure 3.

Stable and unstable magnetizations are easily distinguished by their normalized intensity curves (J/J_0 versus AF peak field; Figure 4), a relationship which was noted after stability had been determined as described above. Specimens with stable magnetizations retain 85–90% or more of their NRM intensity after demagnetization in a peak field of 10 mT. Specimens lacking a stable magnetization retain only about 50–60% of their NRM intensity at the same peak field and show erratic jumps in intensity. Stable specimens recording normal and reversed polarities (i.e., site B) show an intensity drop in early stages of alternating field demagnetization much like that of specimens without a stable magnetization. Intensity increases again after demagnetization in peak fields of about 50 mT.

Specimens from site B have both normal and reversed

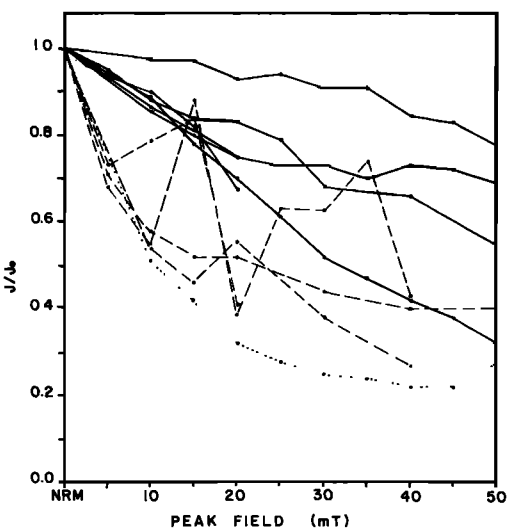


Fig. 4. Plot of normalized intensity (J/J_0) versus AF peak field for samples of BVS pluton. Solid lines show stable magnetizations, dashed lines show unstable magnetizations, and dotted line shows reversed magnetization with normal overprint.

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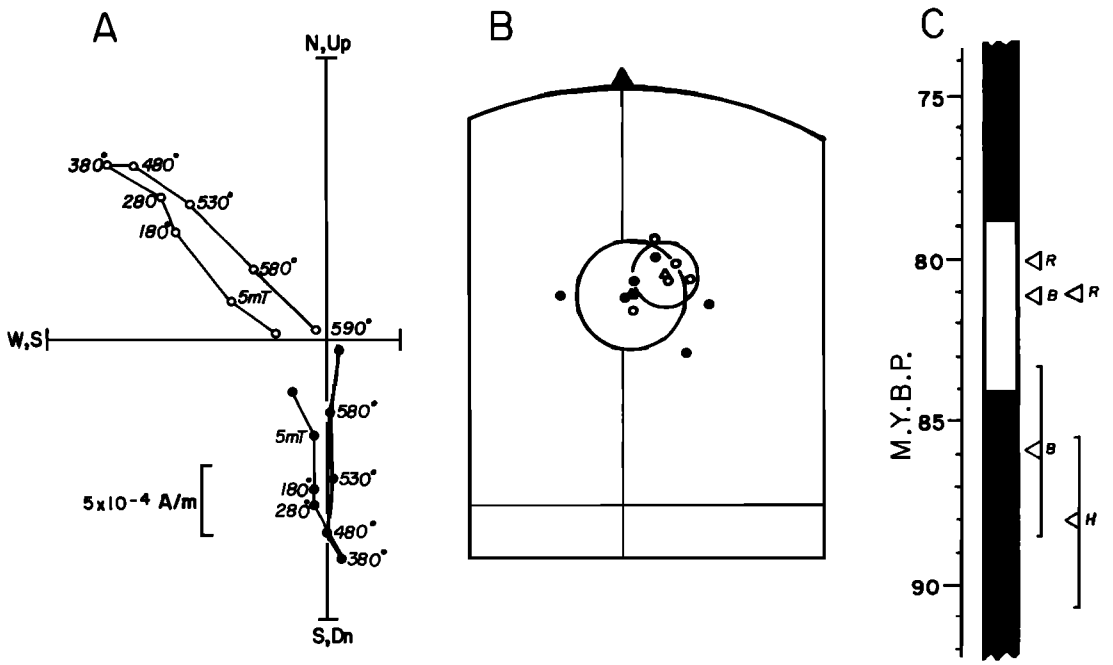


Fig. 5. Antiparallel normal and reversed components of magnetization in samples from site B. (*a*) Thermal demagnetization (°C) orthogonal vector plot. Conventions as in Figure 3. (*b*) Stereonet plot of site B normal directions (solid circles) and reversed (open circles) directions inverted through the origin with respective mean directions and α_{95} circles. (*c*) Reversal time scale of *Lowrie and Alvarez* [1981] for interval of late Cretaceous corresponding to radiometric ages of BVS pluton (shown at right with error bars where available). Normal polarity in black.

components of magnetization which are best separated by thermal demagnetization. The normal component has a lower blocking temperature and dominates the total magnetization vector below about 400°C. The reversed component is seen after demagnetization at temperatures greater than 450° or 500°C. The two components are approximately antiparallel (Figure 5), suggesting that the acquisition of magnetization in these specimens spanned a reversal of the earth's magnetic field and that this change was from reversed to normal polarity. The only known period of reversed polarity between 80 and 90 m.y. ago falls between anomalies 33 and 34 and covers the time period from 84 to 79 m.y. [*Lowrie and Alvarez*, 1981], in good accord with the available isotopic age determinations for the BVS pluton. Because site means are meant to represent only a short time interval, the normal and reversed directions from site B have been averaged separately.

Figure 6*a* shows mean directions for the eight sites in the pluton with stable magnetizations. The mean magnetic direction without tectonic correction is $D=20.9°$, $I=46.4°$ (Table 1), significantly different from present field or axial geocentric dipole field directions for this site. The 80-m.y. paleomagnetic pole for North America [*Irving*, 1979] and its circle of confidence were used to calculate the expected field direction at Tehachapi ($D=337°$, $I=65°$) and its associated oval of confidence. Since the Sierra Nevada as a body has probably moved relative to stable North America because of Basin and Range extension, a similar calculation was performed for the upper Cretaceous Sierra Nevada plutons pole of *Grommé and Merrill* [1965], yielding $D=338°$, $I=66°$, even though it is an older study reporting magnetization directions which may not be completely 'cleaned'. The results from the BVS pluton show a shallow inclination and clockwise declination when compared to either of these reference directions (Table 2).

TECTONIC CORRECTION

Geologic relationships discussed previously indicate that Miocene sediments were deposited on a plutonic surface of significant relief and that relative uplift of at least some plutonic rocks occurred during or after deposition of the Witnet Formation and before deposition of the Kinnick Formation. If Witnet deposition was controlled by an active fault zone, pre-Miocene warping of the Witnet could be a local effect related to motion on that fault. The BVS pluton is nowhere overlain by Witnet Formation so we cannot determine if it was affected by this warping. Correction of the mean direction of the pluton for strike and dip of the Witnet Formation (average N55°E, 58°NW) gives a magnetic vector oriented north and nearly horizontal, far from the expected direction and implying unreasonable northward transport of the BVS pluton which appears to be contiguous with the remainder of the Sierra Nevada batholith. We do not believe this is a valid tectonic indicator and have not used it for the tectonic correction.

The tectonic correction for the BVS pluton comes from attitudes of the Miocene rocks which overlie it. The Kinnick and Bopesta formations are gently folded with dips averaging 10°–15° in the sampling area. Immediately adjacent to faults these dips may steepen to 30° or even 45°, but the effects appear to be very local. About 2 km from sites A and B (Figure 2), Kinnick Formation unconformably overlies the BVS pluton. Four sites were collected in the stratigraphic section exposed here along Highway 58 (J sites) and beds dip about 10° south. Although there is a fault on *Dibblee and Louke's* [1970] map between this locality and the nearest

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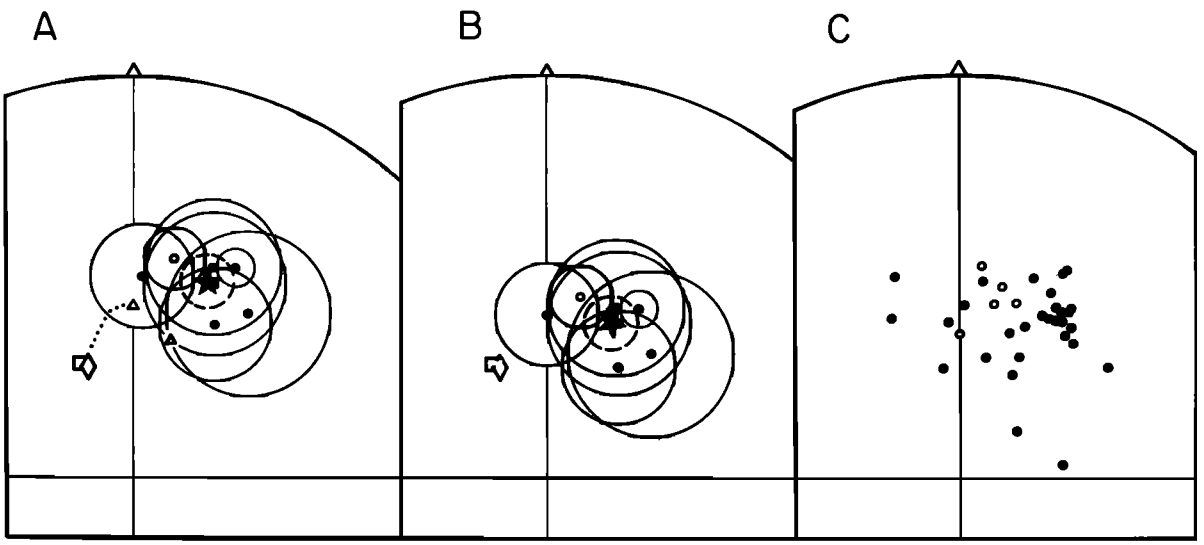


Fig. 6. BVS pluton site means with associated circles of confidence before (Figure 6*a*) and after (Figure 6*b*) tectonic correction. Pluton mean direction shown by star and dashed circle. Open square is direction expected in Tehachapi from pole of *Grommé and Merrill* [1965]; open diamond is direction expected from pole of *Irving* [1979]. Triangles show dipole and present earth field. Dotted line shows path of expected direction from 80 m.y. to present (*c*) Characteristic directions of samples from BVS pluton. Open circles are reversed directions projected through the origin onto the lower hemisphere.

Cretaceous sites, the attitude of the sedimentary rocks appears to be representative of the local tilt, and we have used this as a tectonic correction for the BVS pluton. We believe that the fault is of minor importance since sites on both sides of it as well as beyond its inferred end yield very similar directions. The fault appears to have dip-slip motion, and there is no abrupt change in trend of foliation or metasedimentary rocks across it. Tectonic correction has little effect on declination but steepens the mean inclination, resulting in a mean direction (Table 1) of $D = 22.3°$, $I = 55.7°$, which is still different from the expected directions discussed previously (Figures 6*b* and 6*c*).

Discordant directions can be expressed in terms of rotation (R) and flattening (F), defined by *Beck* [1980]. These values have associated uncertainties ΔR and ΔF, which take into account the uncertainties in both the expected and the observed directions. After tectonic correction the magnetic vector for the BVS pluton shows rotation ($R \pm \Delta R$) of 45° ± 14° and flattening ($F \pm \Delta F$) of 10° ± 9° and 9° ± 8° relative to the poles of *Grommé and Merrill* [1965] and *Irving* [1979], respectively (Table 2).

SECULAR VARIATION

It is difficult to decipher the cooling history of the BVS pluton. It is a large body, presently exposed over more than 1200 km^2, so it probably cooled over several million years. Individual samples of the pluton record both normal and reversed polarities and so were magnetized over tens of thousands of years at least. Another characteristic indication of slow cooling is the single K/Ar sample which yields a greater age for hornblende than for biotite, although the ages overlap statistically. The angular standard deviation expected at Tehachapi using secular variation model D [*Cox*, 1970] is about 11°. If global secular variation was about 15% lower in pre-Tertiary times [*Brock*, 1971], this would lower the expected deviation to 9.3°. The angular standard deviation of the BVS pluton is 7.7° using site statistics and 13.3° using sample statistics. Secular variation is at least partially averaged within sites and probably even within samples. A typical value of k for a site in the pluton is 60 (Table 1) compared to about 250 for a site in basalt (Table 3). Within-site dispersion is nearly as large as between-site dispersion. A two-tier analysis shows that the two values do not differ

TABLE 1. Cretaceous Magnetization Directions

	Location					Corrected					
	λ, °N	φ, °W	Polarity	D, deg	I, deg	D, deg	I, deg	N	R	k	α_{95}
Site											
B_1	35.18	118.48	N	2.6	48.2	358.0	59.4	5	4.922	51.3	10.8
B_2	35.18	118.48	R	10.4	43.0	10.9	52.9	5	4.972	143	6.4
C	35.21	118.54	N	21.3	43.0	22.1	52.0	4	3.930	42.8	14.2
D	35.21	118.54	N	35.5	48.3	40.3	56.5	4	3.894	28.3	17.6
E	35.21	118.54	N	29.0	54.1	33.3	62.7	4	3.949	58.8	12.0
F	35.19	118.53	N	23.2	46.0	24.8	54.5	4	3.919	37.0	15.3
L	35.18	118.52	N	20.1	44.2	21.1	53.2	3	2.834	12.0	37.1
Q	35.20	118.51	N	26.1	41.3	28.5	50.0	5	4.989	364	4.1
Mean				20.9	46.4	22.3	55.7	8	7.928	97.2	5.6

Corrected D and I are declination and inclination after correction for tilt; N is number of samples used in a site; R is length of the vector sum of measurements; $k = (N - 1)/(N - R)$; α_{95} is radius of the circle of 95% confidence about the mean direction.

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TABLE 2. Cretaceous Concordance/Discordance Statistics

	D, deg	I, deg	α/A_{95}	R, deg	$\pm\Delta R$, deg	F, deg	$\pm\Delta F$, deg
BVS pluton (before tectonic correction)	20.9	46.4	5.6				
80-m.y. pole [*Irving*, 1979]*	337	65.1	8	43.9	14.1	18.7	8.3
Sierra Nevada plutons pole [*Gromme and Merrill*, 1965]†	338	65.8	6.6	42.9	14.8	19.4	9.1
BVS pluton (after tectonic correction)	22.3	55.7	5.6				
80-m.y. pole	337	65.1	8	45.3	13.6	9.4	8.3
Sierra Nevada plutons	338	65.8	6.6	44.3	14.4	10.1	9.1

R is positive when observed declination is clockwise of expected declination; F is positive when observed inclination is shallower than expected inclination.
*Direction expected at study site based on pole $\lambda = 69°N$, $\phi = 187°E$.
†Direction expected at study site based on pole $\lambda = 69°N$, $\phi = 195°E$.

significantly at the 97.5% confidence level. We feel secular variation is adequately averaged by the eight sites.

PALEOMAGNETIC RESULTS—MIOCENE ROCKS

Seven sites in basalt flows contributed to the Miocene direction at Tehachapi. Basalt specimens are all of reversed polarity. The more altered sites (M, N, O) require thermal demagnetization to remove completely recent viscous magnetizations. The great circle paths followed during demagnetization originate close to the present dipole field, indicating a two-component magnetization (Figures 7*a* and 7*b*). Fresh specimens show little or no change in direction during demagnetization. Average NRM intensities are about 3×10^{-1} A/m.

Most of the tectonic corrections for the Miocene basalts were obtained from attitudes of nearby sediments and a few from the consistent parting paralleling flow tops and perpendicular to columnar joints. Plunge of folds was removed

TABLE 3. Miocene Magnetization Directions

	Location					Corrected						VGP	
	λ, °N	ϕ, °W	Polarity	D, deg	I, deg	D, deg	I, deg	N	R	k	α_{95}	λ, °N	ϕ, °E
Basalts													
Site													
M	35.31	118.07	R	39.6	33.3	6.9	64.1	4	3.992	375	4.7	78	266
N	35.31	118.07	R	50.2	28.6	30.3	65.5	5	4.970	133	6.7	64	294
O	35.31	118.07	R	44.3	34.8	10.8	68.0	5	4.973	148	6.3	72	265
X	35.17	118.34	R	351.2	60.3	337.2	71.1	5	4.988	333	4.2	64	211
Y	35.17	118.35	R	342.1	41.8	331.3	51.5	8	7.811	37	9.2	66	153
BB	35.15	118.34	R	311.8	57.5	333.7	52.4	6	5.982	278	4.1	68	154
CC	35.14	118.34	R	25.9	67.2	8.7	62.0	8	7.991	778	2.0	79	278
Mean before tectonic correction				18.6	51.5			7	6.264	8.2	22.5		
Mean after tectonic correction						354.5	63.7	7	6.847	39	9.8		
Sedimentary Rocks													
Site													
Z	35.14	118.33	N	12.9	58.9	16.9	66.6	4	3.980	150	7.4	71	278
J3			N/R	357.8	54.0	352.6	62.6	4	3.958	71	11.0	79	211
J5	36.16		N	349.9	51.1	343.9	58.8	4	3.970	100	9.2	76	176
J6	to	118.47	N	345.6	49.3	339.0	56.6	4	3.928	42	14.4	73	165
J7	35.17		N	4.2	61.9	0.2	70.4	4	3.963	81	10.3	70	242
Mean before tectonic correction				359.0	55.4			5	4.949	78	7.7		
Mean after tectonic correction						353.7	63.7	5	4.953	85	8.3		
Mean Miocene						353.7	63.7	12	11.801	55	5.9		
Paleomagnetic pole	78.3	223.2						12	11.58	26	8.6		

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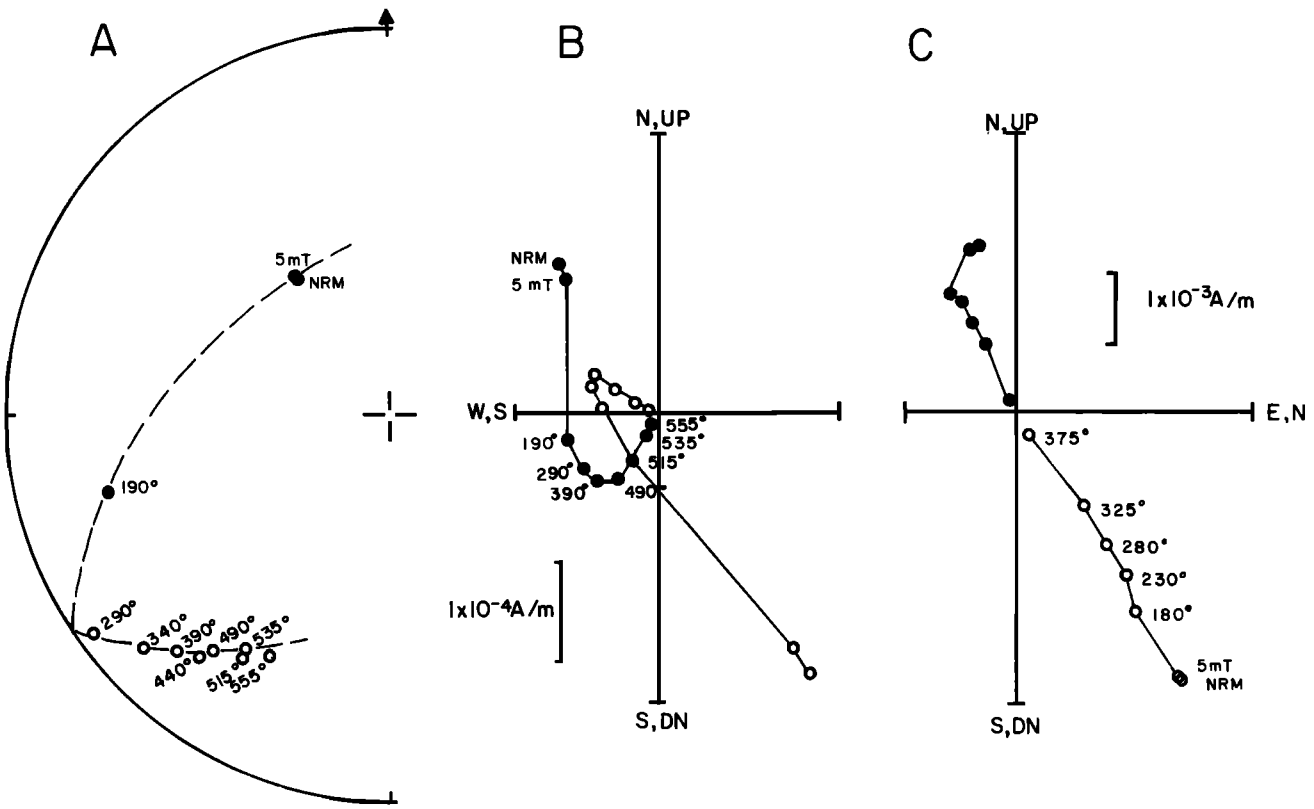


Fig. 7. Magnetic behavior of Miocene rocks. (*a*) Stereonet plot of thermal demagnetization of altered basalt. (*b*) Orthogonal vector plot for same sample. (*c*) Representative orthogonal vector plot for thermal demagnetization of lacustrine rock. Plotting conventions as in Figures 3 and 5.

before beds were rotated to horizontal, although only shallow plunges (~6°) were involved. Tectonically corrected directions show better grouping (k= 39) than uncorrected directions (k= 8.2), an improvement that is significant at the 99% confidence level (Figures 8*a* and 8*b*; Table 3).

Nine sites were collected from tuffs and from a stratigraphic section in lacustrine rocks in the Kinnick Formation. All of these rocks are weakly magnetized, with average NRM intensities of 1×10^{-3} A/m in the tuffs and 1×10^{-4} A/m in lacustrine sediments. Four sites were rejected because they were too weakly magnetized to be measured accurately or they lacked a stable component of magnetization. A few specimens from the tuff and from two sites of lacustrine rocks record two apparently stable magnetizations of opposite polarity with partially overlapping blocking temperature spectra, and it was not possible to isolate completely the reversed magnetization. Rare specimens of lacustrine rocks in the same sites record only a reversed

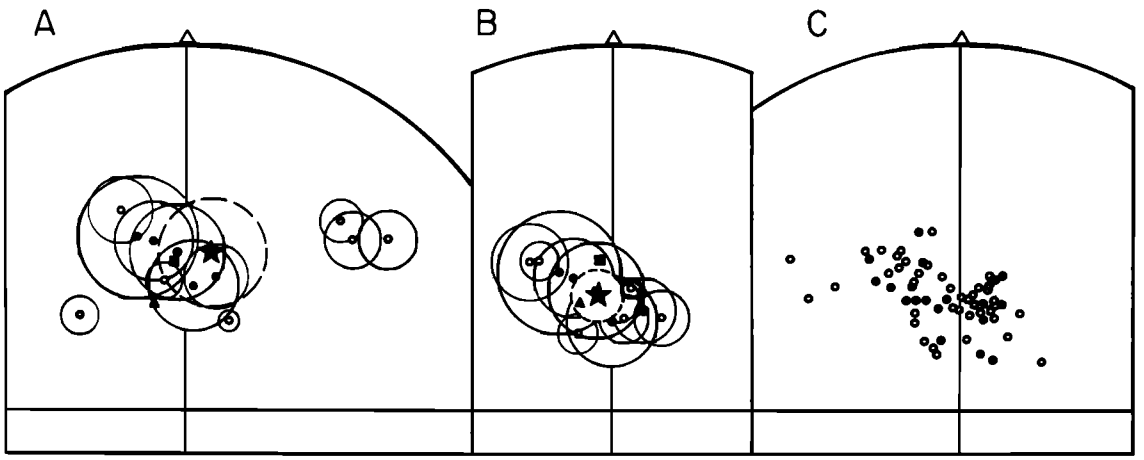


Fig. 8. Site means and associated circles of confidence for Miocene rocks before (Figure 8*a*) and after (Figure 8*b*) tectonic correction. Mean direction shown by star and dashed circle. Square and triangle represent expected directions from poles of *Irving* [1979] and *Van Alstine and de Boer* [1978], respectively. (*c*) Characteristic directions from Miocene rocks. Plotting conventions as in Figure 6.

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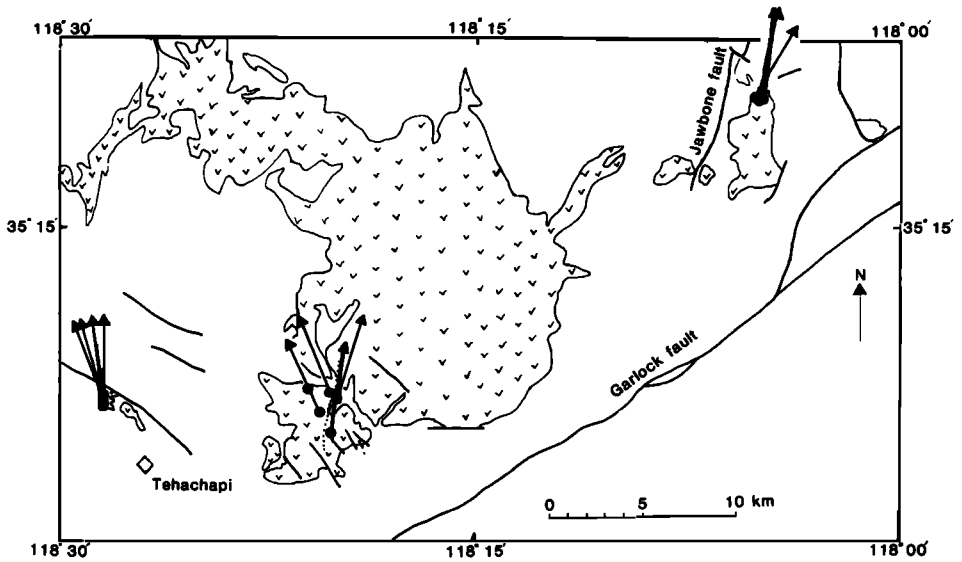


Fig. 9. Miocene sites with magnetic declinations. Possible domain boundary shown by dotted line.

direction which is approximately antiparallel to the normal directions of other samples. There appears to be a correlation between a high degree of silicification of the lacustrine carbonates and reversed direction. The more calcareous specimens, as well as many partially silicified specimens, record only normal polarity (Figure 7c). Magnetic directions from the Kinnick sediments are essentially parallel or antiparallel to directions in the basalts. Figure 8 shows characteristic directions and site means with circles of confidence for all the Miocene rocks. There appears to be some variability of directions in the Miocene rocks. Examining a plot of site declinations on the geologic map (Figure 9) shows that it is possible to divide the sites into two domains with average declinations of about 343° and 17°. We know of no structural or other evidence for a tectonic boundary at this location, however, so we will consider all the Miocene sites together. The mean direction is D=353.7°, I=63.7°.

There are few lower and middle Miocene poles from stable North America. *Irving* [1979] calculates a 20 m.y. pole (λ=87°N, ϕ=167°E) using a sliding window technique which may include several studies on rocks significantly older or younger than the Miocene rocks at Tehachapi. *Van Alstine and de Boer* [1978] calculate a pole position for the period 7–21 m.y. (λ=83°N, ϕ=203°E) using a modal technique. Virtual geomagnetic poles from sites at Tehachapi yield a paleomagnetic pole at λ=78°N, ϕ=223°E (Table 3) which is not significantly different from either of these reference poles. Rotation is −1.3°±12.5° relative to the Van Alstine pole and −3.3°±9.1° relative to the Irving pole (Table 4).

This Miocene study is one of very few in the western United States to show no significant clockwise rotation. It should be useful not only in its implications for the Tehachapi area but in setting spatial and temporal limits on rotated/translated terranes and stable North America. The proximity of unrotated Tehachapi rocks to the strongly clockwise-rotated Miocene rocks of the Transverse Ranges and Channel Islands [*Kamerling and Luyendyk*, 1979] and Oligocene rocks of the southern Coast Ranges [*Greenhaus and Cox*, 1979] emphasizes the degree of fragmentation in southern California and the sharpness of the boundaries between terranes with very different tectonic histories.

INTERPRETATION

Magnetic directions in the upper Cretaceous BVS pluton are divergent from the expected direction for stable North America as well as from the late Cretaceous pole from the Sierra Nevada plutons, and this divergence is almost entirely in declination; the Miocene rocks at Tehachapi, however, show no detectable rotation. Rotation of the BVS pluton must therefore have occurred after magnetization of the pluton but before deposition of the Kinnick Formation, or between about 80 and 20 m.y. B.P.

The mean direction observed for the BVS pluton could be produced by rotation about any number of axes. The horizontal axis which can explain the data is oriented N19°W and requires a rotation of 27°, very similar to the axis requiring the minimum angular rotation (28° about an axis trending N20°W and inclined 26° upward). Because the divergence of magnetic direction in the BVS pluton is nearly all in declination, it could be also the result of a 45° rotation about a nearly vertical axis (Figure 10).

Although there are several ways to explain the results

TABLE 4. Miocene Concordance/Discordance Statistics

	D, deg	I, deg	α/A_{95}	R, deg	$\pm\Delta R$, deg	F, deg	$\pm\Delta F$, deg
This study, Miocene rocks	353.7	63.7	5.9				
North America 20 m.y. [*Irving*, 1979]*	357	56	5	−3.3	9.1	−7.7	7.6
7–21 m.y. [*Van Alstine and de Boer*, 1978]*	355	60	8	−1.3	12.5	−3.7	9.1

*Magnetic direction expected at the study site based on this pole.

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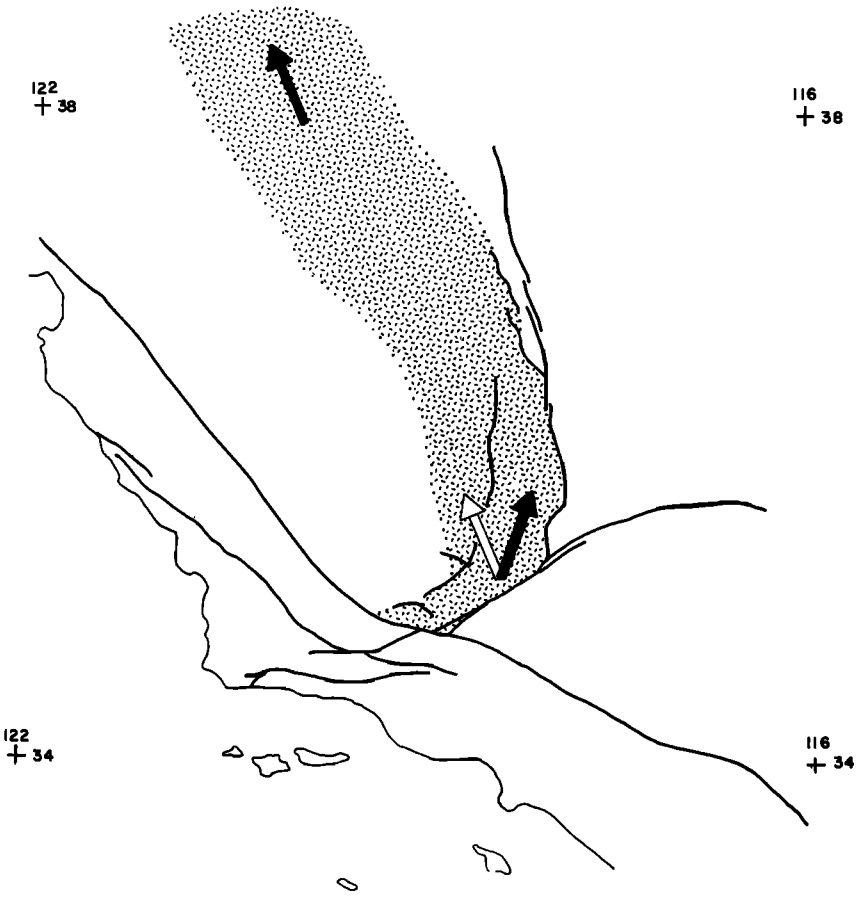


Fig. 10. Central and southern Sierra Nevada batholith (patterned) with paleomagnetic declinations. Single solid arrow is result of *Grommé and Merrill* [1965]; double arrow shows declination observed in BVS pluton (solid) with Grommé and Merrill result (open) for reference.

obtained in this study, we feel that the anomalous magnetic directions of the BVS pluton imply a regional tectonic event. In our preferred model the BVS pluton and the entire southern part of the Sierra Nevada batholith underwent clockwise rotation about a steep axis, probably in Paleocene time. The pluton moved from a position approximately in line with the main body of the Sierra Nevada batholith to its present position.

There are several alternatives to this model.

1. Secular variation has not been adequately averaged. The available results, however, suggest that the samples collected from the BVS pluton average secular variation sufficiently. Miocene volcanic sites alone are not sufficient to average secular variation, but in combination with the fine-grained lacustrine rock sites much more time is sampled. For Miocene as well as Cretaceous rocks, both polarities are present.

2. The BVS pluton is allochthonous to North America and records the magnetic field at its point of origin. This is unlikely because the pluton extends northward across the Breckenridge fault [*Ross,* 1980] into the main body of the Sierran batholith. There are no known post-Cretaceous sutures in the Sierra Nevada. Additionally, the pluton intrudes metavolcanic and metasedimentary terranes which were already juxtaposed at the continental margin [*Ross,* 1980] by the late Cretaceous.

3. The BVS pluton has been remagnetized and records a post-late Cretaceous field. The mean direction of the pluton, however, is unlike field directions for all periods from Cretaceous to Recent (Figure 6*a*). Several methods of age determination do not show the discordance expected to be associated with reheating.

4. The rotation measured is a local effect due to tilt on nearby northwest trending fault(s). This possibility is considered unlikely because although stable sites do not span much distance, they do extend on both sides of a mapped fault as well as beyond its end and no difference in magnetic direction with geographic position is found.

5. The BVS pluton has undergone pre-20 m.y. rotation by itself without accompanying motions of nearby blocks. Given the generally similar structural and magmatic trends to the south and west of the area studied, it seems probable that the deformation we document is part of a larger, regional event.

6. Rotation is produced by distributed right-lateral shear during establishment of the current strike slip regime along the North America–Pacific plate boundary. *Hamilton* [1978] suggests that several right-lateral faults were active along the continental margin in late Oligocene through early to middle Miocene. According to the plate reconstructions of *Engebretson et al.* [1980] the Mendocino triple junction probably passed through the latitude of Tehachapi about 15 to 16 m.y. ago close to or soon after the time of formation of the volcanic rocks we sampled. The passage of the triple junc-

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tion may be related to the volcanism [*Dickinson and Snyder,* 1979]. It is possible that the basement rocks were rotated by distributed shear, then almost immediately covered by the volcanic and lacustrine rocks, but the timing is difficult and the absolute age of the rocks poorly known.

7. Rotation in the Tehachapi region is a result of differential extension to the south or southeast of the Tehachapi area during an earlier (Oligocene-early Miocene?) phase of Basin and Range activity. This would require a greater extension at the latitude of the southernmost Sierra Nevada than further north in order to push the southern end clockwise. The motion must be completed by about the time of deposition of the Kinnick Formation. The spatial and temporal relationships of this earlier phase of Basin and Range extension are not completely understood. In north-central Nevada at least, the extension [*Zoback and Thompson,* 1978] is probably too young. Extension recorded in some of the core complexes and surrounding regions [e.g., *Compton et al.,* 1977; *Reynolds and Rehrig,* 1980; *Gans,* 1981] is closer to the right age. This mechanism cannot be ruled out, but at this time there is a lack of geologic evidence to tie it to the Tehachapi rotation.

It seems most likely that the Tehachapi area is part of a large oroclinal bend affecting to different degrees the entire southern Sierra Nevada. Since it is difficult to distinguish paleomagnetically between ductile folding and brittle fault block motion the observed rotation could result from one or a combination of the two processes.

A considerable amount of evidence exists for deformation in south central and coastal California in the Paleocene [*Dickinson et al.,* 1979; *Graham,* 1976]. Widespread, commonly angular unconformities exist between upper Cretaceous or lower Paleocene and lower Eocene rocks. Franciscan detritus in units just above this unconformity in several localities indicates uplift and erosion of trench fill deposits during this time. Along the east side of the San Andreas fault near Coalinga 'Neogene' wrench folds apparently originated during an episode of Paleogene right-lateral shear, affecting facies patterns in lower Eocene rocks which overlie the Paleocene unconformity [*Harding,* 1976]. In the Temblor Range northwest of Tehachapi the Eocene Point of Rocks Sandstone comprises sediments from a granitic (arc) terrane transported from south to north, in distinct contrast to the east to west transport indicated for the lower and upper Cretaceous sediments it overlies. This relationship has been interpreted to indicate the presence in Eocene time of Salinian granitic rocks to the south [*Clarke and Nilsen,* 1973], but it is also compatible with the presence of rotated Tehachapi/San Emigdio mountains arc rocks by Eocene time. Eocene through Oligocene shorelines all along the north side of the Tehachapi and San Emigdio Mountains trend east-west [*Nilsen,* 1973].

Reconstruction of Kula-Farallon-North America plate motions (D. Engebretson, personal communication, 1981) indicates that the Kula plate was being subducted beneath North America at the latitude of the southern Sierra Nevada very rapidly and in an oblique dextral sense in the late Cretaceous and Paleocene. Both the velocity and degree of obliquity decline markedly by the early Eocene. There is a growing body of geological evidence [e.g., *Nilsen,* 1978] for significant right-lateral strike slip motion along part of the North American Cordillera in latest Cretaceous and Paleocene time. Oroclinal bending of the southernmost Sierra Nevada is one possible result of motion along this proto-San Andreas zone.

Several recent papers address the possibility of rotation in this area. *Ross* [1980] postulates a post-late Cretaceous orocline in the southern Sierra Nevada with a tightly folded and contorted axial zone partially removed by motion on the Garlock fault. Unbending creates generally better alignment of metamorphic and plutonic trends. *Howell* [1975] proposes a bend or bulge in batholithic trends in this area, possibly postdating pluton emplacement, and envisions Salinian basement being sliced off this bulge by late Cretaceous and Paleocene(?) right-lateral faulting. *Burchfiel and Davis* [1981] produce a pre-early Miocene reconstruction that shows a major intraplate Z fold between the Peninsular Ranges and the Sierra Nevada, the short limb of which is formed by the Tehachapi-San Emigdio-Salinian ranges. Several distinct terranes and structural trends of Mesozoic and older ages are involved in this flexure. They feel the bending is complex on a small scale and is post-late Cretaceous and pre-Eocene in age.

The paleomagnetic results from Tehachapi are consistent with rotation of the southern part of the Sierra Nevada from an originally fairly straight NNW trend to a NE and E trending 'hook'. The extent of the rotated terrane is uncertain but the Jurassic Independence dike swarm [*Moore and Hopson,* 1961] of the eastern Sierra Nevada, White-Inyo Mountains, and Mojave desert appears to maintain a fairly constant trend and may provide an eastern limit for the rotation. The western boundary of the rotated terrane has been modified by Neogene motion along the San Andreas fault. Delineation of the original western boundary of the terrane depends on clarification of the relation of Salinian granitic rocks to the Sierra Nevada and the rest of the Cordillera, and their history of motion. The southern boundary of the rotated terrane may be in the western Mojave block, as *Burchfiel and Davis* [1981] suggest. Space problems result if part of the Sierra Nevada rotates with respect to the regions surrounding it. We hope to be able to resolve some of these problems with additional paleomagnetic studies.

Acknowledgments. We are indebted to Eugene Schweig for his able help in the field and to Virginia and Tom Green for their hospitality. We also thank Stephan Graham for useful discussions and Myrl Beck, David Engebretson, James Magill, Elizabeth Miller, Don Ross, and two anonymous reviewers for their comments on drafts of the manuscript. This study was funded in part by grants from the Petroleum Research Fund (PRF11405-AC2), National Science Foundation (EAR77-00120, EAR79-19712), and Shell Companies Foundation to the Geophysics Department and School of Earth Sciences at Stanford.

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(Received May 4, 1981;
revised November 6, 1981;
accepted December 4, 1981.)

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